Methodology — Leakage-Free Detect → Explain → Reveal Pipeline for Text Steganography

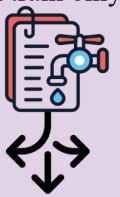
1 Cover corpora
IMDB (train / val / test)
WikiText-2 and SMS Spam
(external validation only)



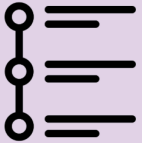
2 Stego generation
ZeroWidth, Whitespace and
Synonym channels + random
payload → 4 labelled classes



3 Leakage-free split
4 variants of a cover share
one split; TF-IDF fitted
on train only



4 Feature extraction
17 interpretable features;
TF-IDF; visible-character
coding for transformers



5 Model training
LightGBM-Lite / -TFIDF, ALBERT,
RoBERTa, Qwen-LoRA; Rule and
TextCNN baselines; multi-seed



6 Evaluation
In-domain + external corpora;
1%-FPR operating points; rate,
normalisation, unseen embedders



7 Explain and reveal
SHAP per alert; payload
decoding; Gradio app, models
and code released

